

## Common Principles and Specific Mechanisms of Mitophagy from Yeast to Humans.

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Mitochondria are double membrane-bound organelles in eukaryotic cells essential to a variety of cellular functions including energy conversion and ATP production, iron-sulfur biogenesis, lipid and amino acid metabolism, and regulating apoptosis and stress responses. Mitochondrial dysfunction is mechanistically linked to several neurodegenerative diseases, cancer, and ageing. Excessive and dysfunctional/damaged mitochondria are degraded by selective autophagic pathways known as mitophagy. Both budding yeast and mammals use the well-conserved machinery of core autophagy-related genes (ATGs) to execute and regulate mitophagy. In mammalian cells, the PINK1-PARKIN mitophagy pathway is a well-studied pathway that senses dysfunctional mitochondria and marks them for degradation in the lysosome. PINK1-PARKIN mediated mitophagy relies on ubiquitin-binding mitophagy adaptors that are non-ATG proteins. Loss-of-function mutations in PINK1 and PARKIN are linked to Parkinson's disease (PD) in humans, and defective mitophagy is proposed to be a main pathomechanism. Despite the common view that yeast cells lack PINK1- and PARKIN-homologs and that mitophagy in yeast is solely regulated by receptor-mediated mitophagy, some studies suggest that a ubiquitination-dependent mitophagy pathway also exists. Here, we will discuss shared mechanisms between mammals and yeast, how mitophagy in the latter is regulated in a ubiquitin-dependent and -independent manner, and why these pathways are essential for yeast cell survival and fitness under various physiological stress conditions.

## Mitochondrial gateways to cancer.

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Mitochondria are required for cellular survival, yet can also orchestrate cell death. The peculiar biochemical properties of these organelles, which are intimately linked to their compartmentalized ultrastructure, provide an optimal microenvironment for multiple biosynthetic and bioenergetic pathways. Most intracellular ATP is generated by mitochondrial respiration, which also represents the most relevant source of intracellular reactive oxygen species. Mitochondria participate in a plethora of anabolic pathways, including cholesterol, cardiolipin, heme and nucleotide biosynthesis. Moreover, mitochondria integrate numerous pro-survival and pro-death signals, thereby exerting a decisive control over several biochemical cascades leading to cell death, in particular the intrinsic pathway of apoptosis. Therefore, it is not surprising that cancer cells often manifest the deregulation of one or several mitochondrial functions. The six classical hallmarks of cancer (i.e., limitless replication, self-provision of proliferative stimuli, insensitivity to antiproliferative signals, disabled apoptosis, sustained angiogenesis, invasiveness/metastatic potential), as well as other common features of tumors (i.e., avoidance of the immune response, enhanced anabolic metabolism, disabled autophagy) may directly or indirectly implicate deregulated mitochondria. In this review, we discuss several mechanisms by which mitochondria can contribute to malignant transformation and tumor progression.

## New roles for mitochondria in cell death in the reperfused myocardium.

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Mitochondria play an important role in regulating the life and death of cells. They provide the cell with

energy via oxidative phosphorylation but can quickly turn into death-promoting organelles in response to stress by disrupting adenosine triphosphate synthesis, releasing pro-death proteins, and producing reactive oxygen species. Due to their high-energy requirement, cardiac myocytes are abundant in mitochondria and as a result, particularly vulnerable to mitochondrial defects. Myocardial ischaemia and reperfusion are associated with mitochondrial dysfunction and cell death. Therefore, future therapies will focus on preserving mitochondrial integrity and function in hopes of minimizing the impact of ischaemia/reperfusion (I/R) injury. It is well established that myocardial I/R activates both necrosis and apoptosis, and that blocking either process reduces the levels of injury. However, recent studies have demonstrated that alterations in mitochondrial dynamics or clearance of mitochondria via autophagy also can contribute to cell death in the myocardium. In this review, we will discuss these new developments and their impact on the role of cardiac mitochondria in cell death following reperfusion in the heart.

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## Efficient and accurate analysis of mitochondrial morphology in a whole cell with a high-voltage electron microscopy.

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Mitochondria in all eukaryotes are essential organelles responsible for adenosine triphosphate synthesis, calcium homeostasis and steroidogenesis. Because the structure and distribution of mitochondria are highly diverse depending on their function and cellular conditions, it is important to develop a rapid and accurate method to assess their morphology. In this study, we visualize whole mitochondria in cultured cells using high-voltage electron microscopy (HVEM). Compared with conventional transmission electron microscopic approaches, the present method does not require thin sectioning and thus requires less time for image acquisition and processing. Furthermore, compared with fluorescence-based light microscopic approaches, our method provides more accurate size information. Thus, we propose that HVEM is a useful tool for rapid and accurate analysis of mitochondrial morphology and distribution in a cell.

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## Mitochondrial Dynamics in Pulmonary Hypertension.

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Mitochondria are essential organelles for energy production, calcium homeostasis, redox signaling, and other cellular responses involved in pulmonary vascular biology and disease processes. Mitochondrial homeostasis depends on a balance in mitochondrial fusion and fission (dynamics). Mitochondrial dynamics are regulated by a viable circadian clock. Hypoxia and nicotine exposure can cause dysfunctions in mitochondrial dynamics, increases in mitochondrial reactive oxygen species generation and calcium concentration, and decreases in ATP production. These mitochondrial changes contribute significantly to pulmonary vascular oxidative stress, inflammatory responses, contractile dysfunction, pathologic remodeling, and eventually pulmonary hypertension. In this review article, therefore, we primarily summarize recent advances in basic, translational, and clinical studies of circadian roles in mitochondrial metabolism in the pulmonary vasculature. This knowledge may not only be crucial to fully understanding the development of pulmonary hypertension, but also greatly help to create new therapeutic strategies for treating this devastating disease and other related pulmonary disorders.

## The Crosstalk between the Gut Microbiota and Mitochondria during Exercise.

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Many physiological changes occur in response to endurance exercise in order to adapt to the increasing energy needs, mitochondria biogenesis, increased reactive oxygen species (ROS) production and acute inflammatory responses. Mitochondria are organelles within each cell that are crucial for ATP production and are also a major producer of ROS and reactive nitrogen species during intense exercise. Recent evidence shows there is a bidirectional interaction between mitochondria and microbiota. The gut microbiota have been shown to regulate key transcriptional co-activators, transcription factors and enzymes involved in mitochondrial biogenesis such as PGC-1 $\alpha$ , SIRT1, and AMPK genes. Furthermore, the gut microbiota and its metabolites, such as short chain fatty acids and secondary bile acids, also contribute to host energy production, ROS modulation and inflammation in the gut by attenuating TNF $\alpha$ -mediated immune responses and inflammasomes such as NLRP3. On the other hand, mitochondria, particularly mitochondrial ROS production, have a crucial role in regulating the gut microbiota via modulating intestinal barrier function and mucosal immune responses. Recently, it has also been shown that genetic variants within the mitochondrial genome, could affect mitochondrial function and therefore the intestinal microbiota composition and activity. Diet is also known to dramatically modulate the composition of the gut microbiota. Therefore, studies targeting the gut microbiota can be useful for managing mitochondrial related ROS production, pro-inflammatory signals and metabolic limits in endurance athletes.

## A novel mechanism involved in the coupling of mitochondrial biogenesis to oxidative phosphorylation.

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Mitochondria are essential organelles that are central to a multitude of cellular processes, including oxidative phosphorylation (OXPHOS), which produces most of the ATP in animal cells. Thus it is important to understand not only the mechanisms and biogenesis of this energy production machinery but also how it is regulated in both physiological and pathological contexts. A recent study by Ostojić et al. [Cell Metabolism (2013) 18, 567-577] has uncovered a regulatory loop by which the biogenesis of a major enzyme of the OXPHOS pathway, the respiratory complex III, is coupled to the energy producing activity of the mitochondria.

## Mitochondrial Mg(2+) homeostasis decides cellular energy metabolism and vulnerability to stress.

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Cellular energy production processes are composed of many Mg(2+) dependent enzymatic reactions. In fact, dysregulation of Mg(2+) homeostasis is involved in various cellular malfunctions and diseases. Recently, mitochondria, energy-producing organelles, have been known as major intracellular Mg(2+) stores. Several biological stimuli alter mitochondrial Mg(2+) concentration by intracellular redistribution. However, in living cells, whether mitochondrial Mg(2+) alteration affect cellular energy metabolism remains unclear. Mg(2+) transporter of mitochondrial inner membrane MRS2 is an essential component

of mitochondrial Mg(2+) uptake system. Here, we comprehensively analyzed intracellular Mg(2+) levels and energy metabolism in Mrs2 knockdown (KD) cells using fluorescence imaging and metabolome analysis. Dysregulation of mitochondrial Mg(2+) homeostasis disrupted ATP production via shift of mitochondrial energy metabolism and morphology. Moreover, Mrs2 KD sensitized cellular tolerance against cellular stress. These results indicate regulation of mitochondrial Mg(2+) via MRS2 critically decides cellular energy status and cell vulnerability via regulation of mitochondrial Mg(2+) level in response to physiological stimuli.

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## Non-mitochondrial ATP transport.

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Exchange of organelle ATP with cytosolic ADP through the ADP/ATP carrier is a well-characterized feature of mitochondrial metabolism. Obligate intracellular bacteria, such as *Rickettsia prowazekii*, and higher-plant plastids possess another type of adenylate transporter, which exchanges bacterial or plastidic ADP for ATP from the eukaryotic (host cell) cytoplasm. The bacterial and plastidic transporters are similar but do not share significant sequence similarities with the mitochondrial carrier. Recent molecular and biochemical studies are providing deeper insight into the functional and evolutionary relationships between the bacterial and the plant transport proteins.

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## The In Vitro Cytotoxic Effect of Elesclomol on Breast Adenocarcinoma Cells Is Enhanced by Concurrent Treatment with Glycolytic Inhibitors.

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**Background/Objectives:** Glycolysis and mitochondrial oxidative phosphorylation are the two major metabolic pathways for cellular ATP production. The metabolic plasticity displayed by cancer cells allows them to effectively shift between each of these pathways as a means of adapting to various growth conditions, thus ensuring their survival, proliferation and disease progression. Metabolic plasticity also provides cancer cells with the ability to circumvent many traditional monotherapies aimed at only one or the other of the major ATP-producing pathways. The purpose of this study was to determine the effectiveness of a dual treatment strategy aimed simultaneously at both pathways of ATP production in human breast cancer cells. It was hypothesized that concurrent exposure of these cells to the mitochondria-targeting chemotherapeutic agent, elesclomol, in combination with either of two glycolytic inhibitors, 2-deoxy-D-glucose or 3-bromopyruvate, would yield greater in vitro anticancer effects than those observed for any of the compounds used as a single agent. **Methods:** Cytotoxicity and clonogenic assays were employed to assess the survival and proliferation of MCF7 and MDA-MB-231 human breast adenocarcinoma cells exposed to the compounds alone and in combination. **Results:** The data obtained show that the cancer-cell-killing and antiproliferative effects of the dual treatment were significantly enhanced compared to those observed for any of the compounds alone. **Conclusions:** The results of this study are important in that they suggest the possibility of a novel and effective chemotherapeutic strategy for breast cancer cell killing.